

The Impact of Social Media Use on Political Participation

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1. Introduction

In the months leading up to the 2024 U.S. presidential election, Elon Musk leveraged his ownership of X (formerly Twitter) in unprecedented ways: spending over \$277 million to support Donald Trump’s campaign, hosting high-profile interviews with candidates, and implementing daily \$1 million giveaways to registered voters in swing states. While debates about the propriety of such actions dominated headlines, a more fundamental question remained unresolved: does social media usage actually affect whether Americans vote? Despite widespread concern about social media’s influence on democratic politics, empirical evidence on its relationship with voter turnout remains limited and contested, with most existing studies either treating “social media” as an undifferentiated category or examining single platforms in isolation.

The relationship between social media and political participation remains contested. Experimental evidence suggests mobilization effects. Bond et al. (2012) find

that Facebook’s ”I Voted” feature increased turnout by approximately 340,000 voters in 2010, while Fujiwara, Müller, and Schwarz (2024) document similar effects for Twitter in India. However, Allcott et al. (2020) find that paying users to deactivate Facebook for four weeks produced only modest changes in political engagement. Moreover, Boxell, Gentzkow, and Shapiro (2017) argue that social media’s role in political polarization may be overstated, noting that polarization has increased most among older Americans, who are least likely to use such platforms.

A notable limitation of existing work is the treatment of ”social media” as either monolithic or the focus on individual platforms in isolation, typically Facebook. There exists heterogeneity across platforms in user demographics, content moderation, and algorithmic design. Furthermore, the social media landscape has evolved substantially since most existing studies: platforms have shifted toward algorithmic content curation, and platform ownership has become increasingly politicized, most notably with Musk’s 2022 acquisition of Twitter and its re-branding as X.

This paper reexamines the relationship between social media use and voter turnout by exploiting individual-level variation in Facebook and X (formerly Twitter) usage. Unlike Boxell et al. (2017), who use age cohorts to proxy for social media exposure, I leverage direct measures of platform-specific usage from ANES survey data in the 2020 and 2024 presidential elections. The results suggest that X usage is associated with a small but statistically significant positive effect on voter turnout, while Facebook usage shows no significant relationship. These findings suggest that social media’s political influence may be platform-specific, warranting further explanation at a more granular level.

2. Data

I use data from the American National Election Studies (ANES) Time Series surveys conducted during the 2020 and 2024 presidential elections. The ANES uses a multi-mode design combining in-person, telephone, and online interviews. Each

survey consists of two waves: a pre-election wave carried out in September and October immediately preceding an election, and a post-election wave in November and December. Respondents are asked a variety of questions regarding their political habits (voter registration, intention to vote, and voting decisions), political opinions, and demographic information.

The outcome variable of interest is voting in presidential elections, defined as a boolean.

Independent variables of interest are the use (as a boolean) and frequency of use regarding Facebook and X. Facebook and X use is recorded both as a binary variable and as an ordinal variable describing the frequency of use. Ordinal responses are: (0) Not at all, (1) Many times a day, (2) A few times a day, (3) Once a day, (4) A few times each week, (5) About once a week, (6) Once or twice a month, (7) Less than once a month. For the purposes of regression, these are reduced to four categories: (0) Not at all, (1) Infrequently (less than weekly), (2) Weekly, (3) Daily or more.

Control variables are related to race, age, state of residence, sex, and educational attainment.

The combined sample includes 8,947 respondents: 5,518 from 2020 and 3,429 from 2024. 75% of the sample use Facebook in any capacity, compared to just 30% of the sample using X. Individuals in this sample demonstrate high levels of political engagement. The sample exhibits substantially higher political engagement than the general population. Only 2.0% of respondents (184 individuals) were unregistered before the election, and only 1.2% (108 individuals) reported not voting—compared to approximately 34% abstention in the voting-eligible population during recent presidential elections.

2.1. Sample Selection and Limitations

The high participation rates in this sample reflect substantial positive selection for survey participation. Approximately 99% of respondents both registered and voted, compared to roughly 66% turnout among the voting-eligible population in recent presidential elections. This selection poses two key challenges for interpretation.

First, individuals who agree to participate in lengthy political surveys are likely more politically engaged than non-participants, which limits the external validity of the findings. The estimated associations between social media use and turnout may not generalize to less engaged populations.

Second, the limited variation in the outcome variable (only 1.2% abstention) reduces statistical power to detect effects. Even substantively meaningful relationships between platform usage and turnout may be difficult to identify in a sample where nearly everyone votes, regardless of social media use.

Additionally, survey participation may temporarily increase political engagement through priming effects, potentially inducing a positive correlation between survey responses and voting, even in the absence of any effect from social media.

Social media use differs substantially across platforms Facebook use (75.1%) is far more popular than X use (30.8%). Additionally, Facebook users reported using Facebook more regularly than X users used X. These differences reflect both Facebook’s broader demographic reach and X’s more specialized user base, which is concentrated among politically attentive individuals and journalists.

3. Methods

I estimate the effect of social media on voting using two methods. The first model utilizes a logit regression to estimate the likelihood being registered to vote before the election given social media use. I then estimate the likelihood of voting given Facebook and X use with an OLS and logit model. The initial logit regression is

expanded to the final form:

$$\text{logit}(\Pr(\text{Voted}_i = 1)) = \beta_0 + \beta_1 \cdot \text{FBFreq}_i + \beta_2 \cdot \text{XFreq}_i + \beta_3 \cdot \text{Year}_i + \mathbf{X}'_i \boldsymbol{\beta} + \varepsilon_i \quad (1)$$

where $\Pr(\Pr(\text{Voted}_i = 1))$ represents the likelihood of voting, *FBFreq* represents an ordinal measure of Facebook use and *XFreq* an ordinal measure of X use. Co-variates represented by $\mathbf{X}'_i \boldsymbol{\beta}$ include political variables related to party affiliation and an ordinal measure of political attentiveness. Demographic covariates included are race, age, state of residence, sex, and education achieved.

I also estimate a logit model on the likelihood of voting including instead an interaction of a binary measure of Facebook/X use and the election year.

$$\begin{aligned} \text{logit}(\Pr(\text{Voted}_i = 1)) = & \beta_0 + \beta_1 \cdot \text{FBUse}_i + \beta_2 \cdot \text{XUse}_i + \beta_3 \cdot \text{Year}_i \\ & + \beta_4 \cdot (\text{FBUse}_i \times \text{Year}_i) + \beta_5 \cdot (\text{XUse}_i \times \text{Year}_i) + \mathbf{X}'_i \boldsymbol{\beta} + \varepsilon_i \quad (2) \end{aligned}$$

4. Results

Table 4 presents average marginal effects from logit models estimating the relationship between social media usage and voter turnout. The results reveal notable heterogeneity across platforms. X users are 1.7 percentage points more likely to vote than non-users ($p=0.038$), while Facebook usage shows no significant relationship with turnout (0.2pp, $p=0.265$).

These effect sizes, while modest in absolute terms, are meaningful given the limited variation in the outcome variable. The effect X is equivalent to approximately 1.7% of the baseline turnout rate, suggesting that the use of the platform correlates with political participation at the margin even in a sample with near-

universal voting.

Despite Facebook’s substantially larger user base (75% of the sample) compared to X (31%), only X usage predicts turnout. This heterogeneity suggests that social media’s relationship with political participation depends on platform-specific characteristics rather than social media use in general.

Table 5 examines whether the relationship between platform usage and turnout varies by usage frequency. The results reveal a non-linear pattern concentrated among moderate users. For X, infrequent users (less than weekly) are 2.0 percentage points more likely to vote than non-users ($p < 0.001$), and weekly users show a similar 1.8 percentage point effect ($p < 0.001$). However, daily users show no significant relationship (0.4pp, $p = 0.699$).

Facebook exhibits a similar but weaker pattern. Infrequent users show a marginally significant 1.3 percentage point effect ($p = 0.018$), but weekly and daily users show no significant differences from non-users. Joint significance tests confirm that X frequency categories are collectively significant ($\chi^2 = 14.03$, $p = 0.003$), while Facebook categories are not ($\chi^2 = 5.34$, $p = 0.148$).

This inverted-U relationship is consistent with two explanations. First, heavy platform users may represent a highly selected group already at the participation ceiling, leaving little room for additional mobilization. Second, moderate exposure may provide optimal information benefits without the saturation or time displacement effects of heavy use. Distinguishing these mechanisms would require data on actual content exposure and political engagement prior to platform adoption, which are unavailable in the ANES.

To test whether the relationship between platform usage and turnout changed following Elon Musk’s 2022 acquisition and subsequent politicization of X, I estimate models including interactions between platform usage and election year. The results reveal remarkable temporal stability: X users were 1.5 percentage points

more likely to vote in 2020 ($p=0.041$) and 1.3 percentage points more likely in 2024 ($p=0.038$), with the 0.2 percentage point difference not statistically significant ($\chi^2=0.05$, $p=0.817$). Facebook similarly shows no significant effects in either year and no temporal change ($\chi^2=2.30$, $p=0.129$).

Despite Musk’s \$277 million spending on the Trump campaign, purported daily \$1 million voter giveaways, and visible political involvement throughout 2024, the relationship between X usage and turnout remained unchanged. This stability suggests that platform effects on political participation may be more deeply rooted in user composition and platform characteristics than in ownership changes or high-profile political interventions. The association between X usage and voting appears to reflect selection of politically engaged individuals onto the platform rather than mobilization effects of platform ownership or content.

As expected given 98% pre-election registration rates, platform usage shows no significant relationship with voter registration status. These null results reflect the limited variation in registration rather than the absence of platform effects on political behavior more generally.

5. Conclusion

This paper examines the relationship between social media usage and voter turnout using data from the 2020 and 2024 presidential elections. Three main findings emerge. First, social media’s association with political participation is platform-specific: X users are approximately 1.5 percentage points more likely to vote than non-users, while Facebook usage shows no significant relationship. Second, the X effect is non-linear, concentrated among infrequent and weekly users rather than daily users, suggesting either ceiling effects among heavy users or diminishing returns to platform exposure. Third, despite Elon Musk’s 2022 acquisition of X and his subsequent \$277 million spending on the 2024 campaign, the relationship between X usage and turnout remained stable across both elections.

These patterns are consistent with selection into platform usage rather than platform-driven mobilization. X users may be more likely to vote not because the platform mobilizes them, but because politically engaged individuals disproportionately select onto X. This interpretation is supported by the temporal stability finding: if Musk’s political interventions in 2024 mobilized voters, we would expect to observe a strengthened relationship between X usage and turnout. Instead, the effect remained unchanged, suggesting that platform characteristics and user composition established before the ownership change continue to drive the observed associations.

The concentration of effects among moderate users warrants further investigation. Exploratory analysis of political posting behavior suggests that the relationship between platform engagement and turnout may follow an inverted-U pattern, with occasional political posters showing stronger associations than passive consumers or frequent posters. However, interpreting these patterns is complicated by severe endogeneity: political posting is itself a form of political participation, making it difficult to distinguish whether posting drives voting or whether both reflect underlying political engagement. Future research employing experimental or quasi-experimental designs would be needed to establish causal relationships and distinguish selection from treatment effects.

Although intriguing, these findings suffer significant limitations. These findings reflect conditional correlations rather than causal effects. Individuals self-select into platform usage based on political interest, making it impossible to distinguish whether platform usage increases turnout or whether politically engaged individuals are simply more likely to use certain platforms. Temporal stability findings argue against Musk-era mobilization effects, but do not rule out selection mechanisms operating in both periods. This analysis is also limited to presidential elections, where turnout is highest and information environments are saturated. The relationship between social media usage and participation may differ substantially in lower-salience elections where information costs are higher and mobilization effects may be more

pronounced.

Experimental or quasi-experimental designs such as platform bans, algorithm changes, or randomized content exposure could provide causal evidence on whether social media affects political participation or merely reflects selection. Extending this analysis to other platforms, particularly TikTok and Instagram, would clarify whether the platform heterogeneity observed here generalizes to newer social media environments with different user demographics and content characteristics. Future work could also potentially examine younger, less politically engaged populations where the registration-to-voting pipeline may reveal platform effects that are obscured in samples of already-registered voters. The two-stage process of voter registration and turnout may be more meaningfully examined among populations with lower baseline registration rates, such as young adults or recently naturalized citizens, where social media might play a more substantial role in political mobilization.

Despite these limitations, the findings contribute to understanding social media’s role in political participation by demonstrating substantial platform heterogeneity, identifying non-linear relationships between usage intensity and turnout, and showing that high-profile ownership changes do not necessarily alter established platform-user relationships. Social media’s influence on democratic politics appears to depend on which platforms are examined and how users engage with them, rather than on “social media” as an undifferentiated category.

Table 1: Sample Demographics

	(1) N	Mean	SD	Min	Max
Age (years)	8947	53.74	16.79	18	80
Female	8947	0.54	0.50	0	1
Education Level (1-8)	8947	4.81	1.99	1	8

Sample: 8,947 respondents from ANES 2020 and 2024 (2020: 5,518; 2024: 3,429).
Female is binary indicator (1=female, 0=male).
Education: 1=less than high school, 8=professional/doctoral degree.
Mean education (4.81) corresponds to some college.
Race/Ethnicity: 76.7% White, 11.8% Black, 6.2% Hispanic, 3.5% Asian, 1.9% Other.

Table 2: Political Attitudes and Engagement

	(1) N	Mean	SD	Min	Max
Party Identification (1-7)	8947	3.83	2.34	1	7
Democratic Party Thermometer (0-100)	8947	46.15	33.11	-8	100
Republican Party Thermometer (0-100)	8947	42.50	33.05	-8	100
Affective Polarization	8947	54.37	29.00	0	100
Political Attention (1-5)	8947	2.22	0.99	1	5

Sample: 8,947 respondents from ANES 2020 and 2024.

Party ID: 1=Strong Democrat, 4=Independent, 7=Strong Republican.

Mean party ID (3.84) indicates slight Democratic lean in sample.

Thermometers: 0=coldest feelings, 100=warmest feelings toward party.

Affective polarization: absolute difference between party thermometers.

Political attention: 1=always pay attention, 5=never. Mean (2.22) = most of the time.

Table 3: Political Participation and Social Media Usage

	(1) N	Mean	SD	Min	Max
<i>Outcome Variable</i>					
Voted in Presidential Election	8947	0.99	0.11	0	1
<i>Social Media Usage</i>					
Uses Facebook	8947	0.75	0.43	0	1
Uses X (Twitter)	8947	0.31	0.46	0	1
Facebook Usage Frequency (0-3)	8947	1.91	1.26	0	3
X Usage Frequency (0-3)	8947	0.58	0.99	0	3

Sample: 8,947 respondents from ANES 2020 and 2024.

Voted and platform usage are binary indicators.

98.8% of sample voted (N=8,985), compared to 66% national turnout.

75.1% use Facebook; 30.8% use X.

Usage frequency categories: 0=not at all, 1=infrequent (less than weekly),

2=weekly (1-4 times/week), 3=daily or more (once+ per day).

Mean FB frequency (1.90) = between infrequent and weekly.

Mean X frequency (0.58) = between non-user and infrequent.

Table 4: Effect of Social Media Usage on Voting (Average Marginal Effects)

	(1) Base	(2) Political	(3) Full
Facebook User	0.001 (0.004)	0.001 (0.005)	0.002 (0.005)
X User	0.014** (0.006)	0.014** (0.007)	0.017** (0.008)
Observations	8,947	8,947	7,255

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Base model includes year fixed effect.

Political controls add political attention and party identification.

Full controls add education, race, state fixed effects, age, and sex.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Social Media Usage Frequency and Voter Turnout

	(1) Political Controls	(2) Full Controls
Facebook: Infrequent	0.011** (0.005)	0.013** (0.006)
Facebook: Weekly	0.005 (0.006)	0.007 (0.006)
Facebook: Daily+	-0.002 (0.005)	-0.001 (0.006)
X: Infrequent	0.016*** (0.003)	0.020*** (0.004)
X: Weekly	0.015*** (0.004)	0.018*** (0.005)
X: Daily+	0.004 (0.009)	0.004 (0.011)
Observations	8,947	7,255

Average marginal effects from logit models.

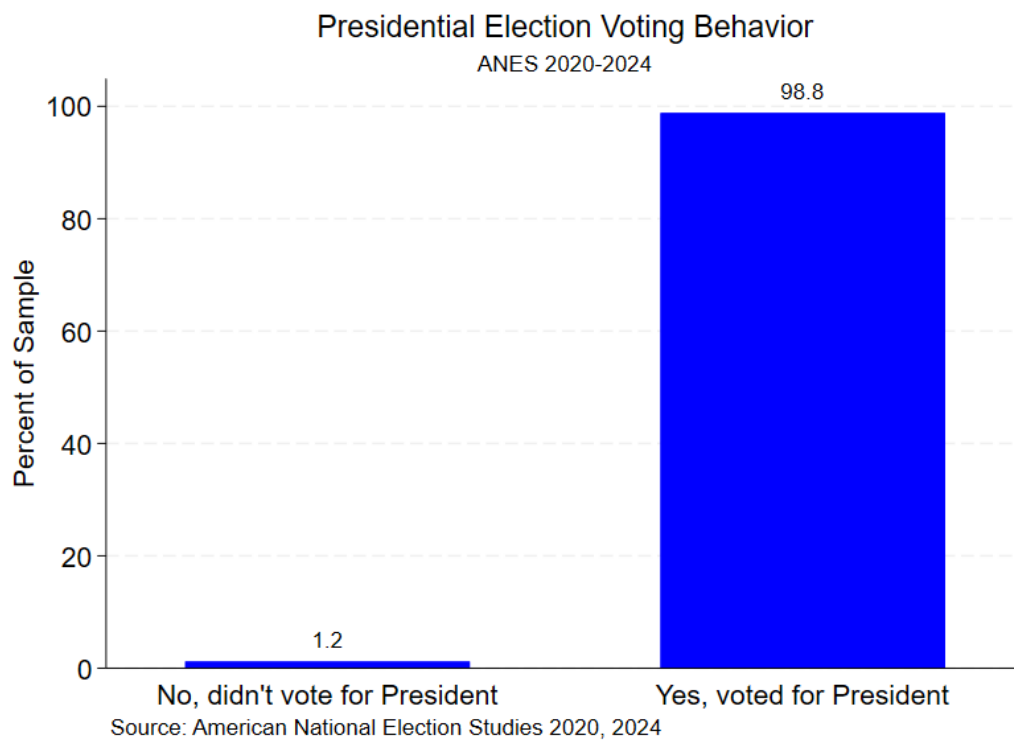
Reference category: Non-users of each platform.

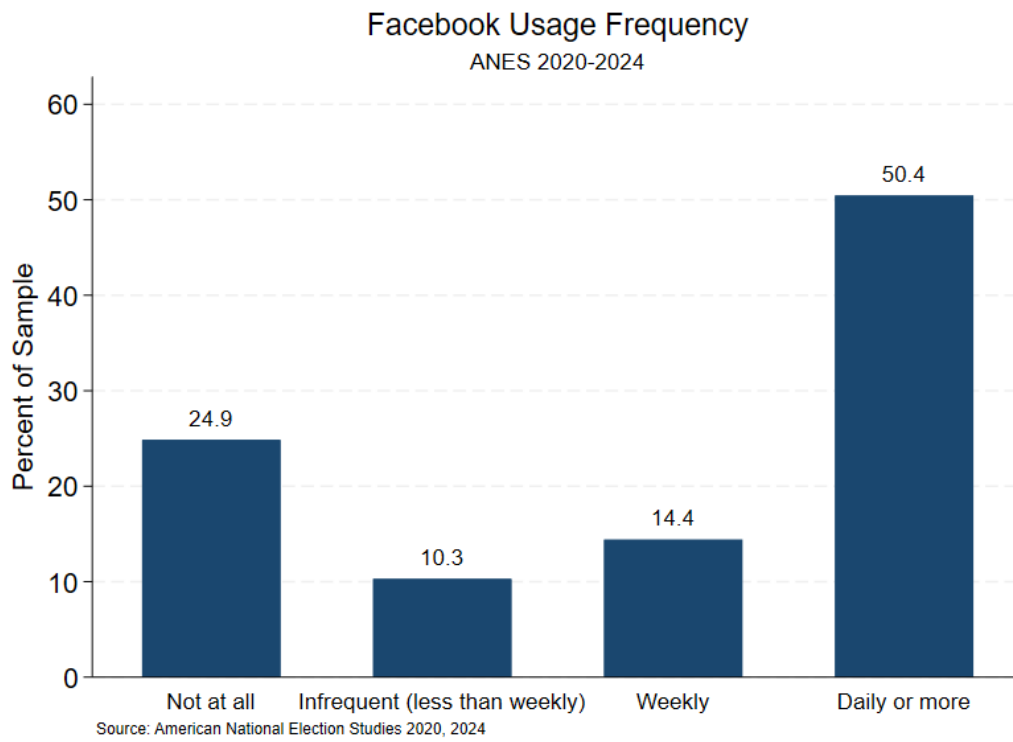
Infrequent = less than weekly; Weekly = 1-4 times/week; Daily+ = once or more per day.

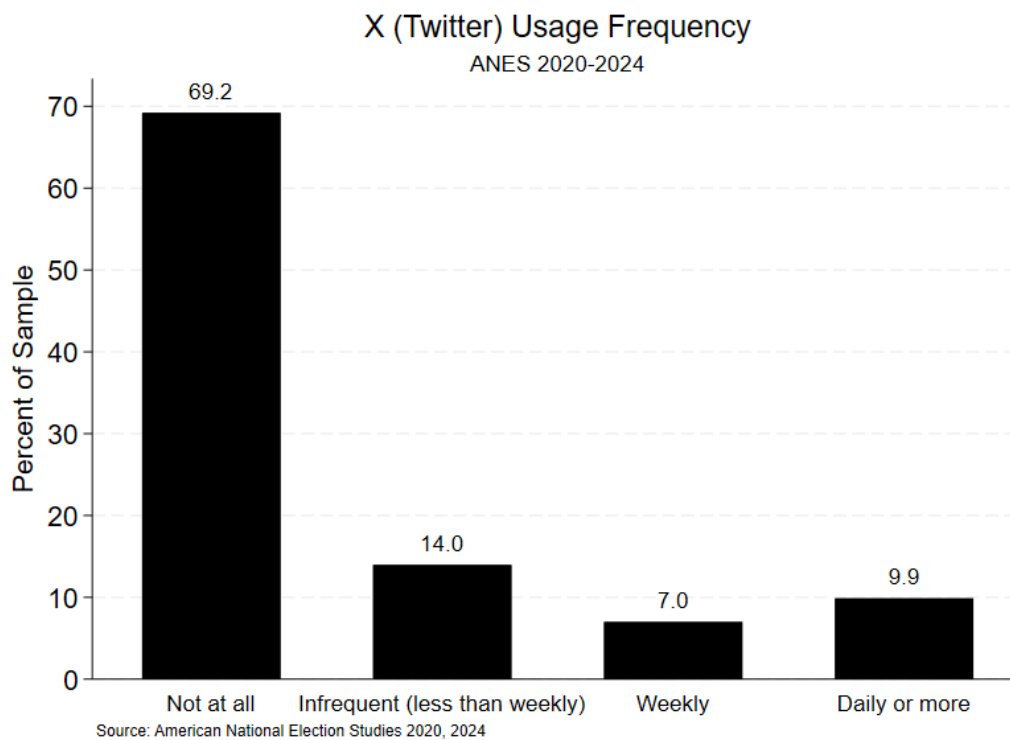
Political Controls: year FE, political attention, party identification.

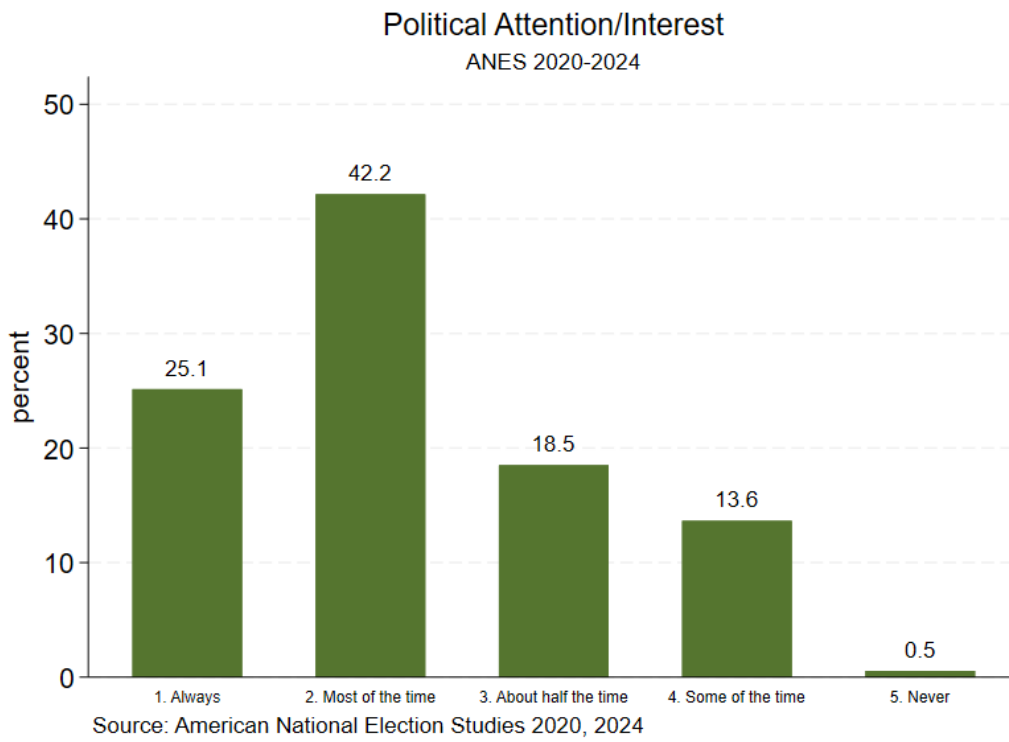
Full Controls: adds age, sex, education, race, and state fixed effects.

Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01









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